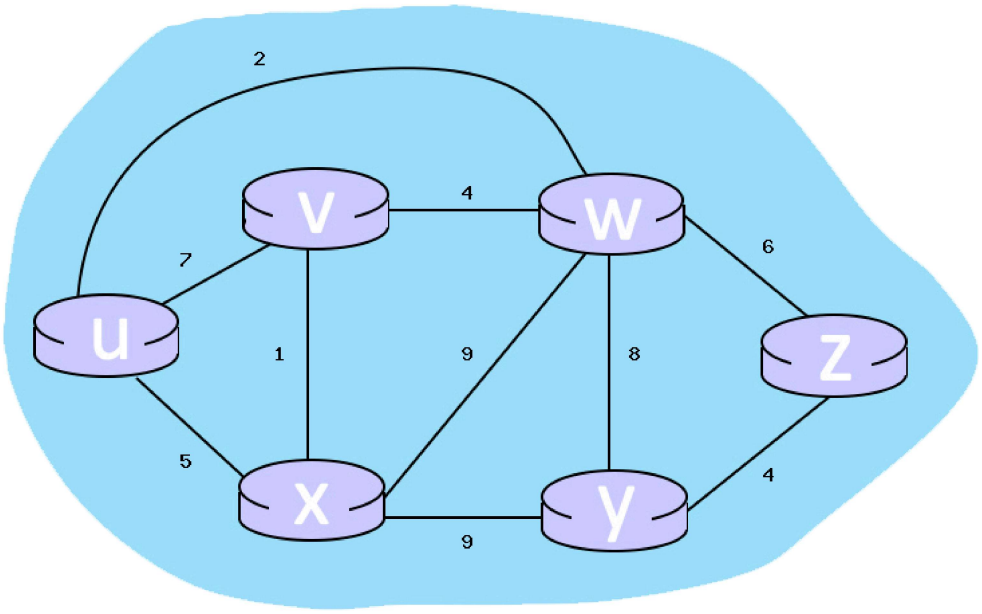
INTERACTIVE END-OF-CHAPTER EXERCISES

DIJKSTRA'S LINK STATE ALGORITHM (FOR COMPUTING LEAST COST PATHS)

Consider the 6-node network shown below, with the given link costs.



Using Dijkstra's algorithm, find the least cost path from source node U to all other destinations and answer the following questions

QUESTION LIST

1. What is the shortest distance to node z and what node is its predecessor? Write your answer as n,p
2. What is the shortest distance to node y and what node is its predecessor? Write your answer as n,p
3. What is the shortest distance to node w and what node is its predecessor? Write your answer as n,p

SOLUTION

1. The minimum distance from node u to node z is 8, and node z's predecessor is node w. The full answer was: 8,w
2. The minimum distance from node u to node y is 10, and node y's predecessor is node w. The full answer was: 10,w
3. The minimum distance from node u to node w is 2, and node w's predecessor is node u. The full answer was: 2,u

QUESTION 1 OF 3

What is the shortest distance to node z and what node is its predecessor? Write your answer as n,p

Answer

CHECK HINT SKIP

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